Code: 20A03703a **R20**

B.Tech IV Year I Semester (R20) Regular & Supplementary Examinations December/January 2024/2025

MECHATRONICS & MEMS

(Mechanical Engineering)

Time: 3 hours Max. Marks: 70

PART – A

(Compulsory Question)

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1 Answer the following: (10 X 02 = 20 Marks)

	(a)	‘CNC machine is a mechatronic product’-justify.	2M
	(b)	List the key elements of a mechatronic system.	2M
	(c)	Define sensitivity of a sensor.	2M
	(d)	List any two applications where nano sensors are used.	2M
	(e)	Distinguish between AC and DC motors with respect to speed control and efficiency.	2M
	(f)	Define Piezoelectric effect.	2M
	(g)	What is the purpose of program memory (Flash memory) and data memory (RAM) in a microcontroller?	2M
	(h)	List the factors to be considered in selection of controllers.	2M
	(i)	Why is it essential to understand the effect of scaling in MEMS?	2M
	(j)	What is Lab on chip?	2M

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

2		With a flowchart, explain the mechatronic design process.	10M
		OR	
3		With a schematic diagram, brief on the sensors and actuators used in a typical automatic inspection system.	10M
4		The force exerted in the wrist of a robot is to be monitored continuously so as to keep the force within control limits. Suggest a suitable sensor for the above application and justify the same. With a neat sketch, explain the construction and working of the suggested sensor.	10M
		OR	
5		With suitable sketches, explain any two types of flow sensors.	10M
6	(a)	Compare electrical, hydraulic and pneumatic systems with respect to payload, power-to-weight ratio, cost and accuracy.	6M
	(b)	Brief on the selection criteria for actuators.	4M
		OR	
7		Describe about shape memory alloys, highlighting on the shape memory effect.	10M
8		With a neat sketch, explain the architecture of 8085 microprocessor.	10M
		OR	
9	(a)	With a suitable example, illustrate latching in a PLC.	5M
	(b)	Explain on-delay and off-delay timers using suitable examples.	5M
10		Illustrate the process of depositing thin films on a substrate using Physical Vapour Deposition.	10M
		OR	
11		Illustrate how high-aspect microstructures are fabricated using LIGA process.	10M

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